

SkillForge

AI Agent Skills Marketplace

Product Plan · Tech Stack · Action Roadmap

1. Vision & Concept

SkillForge is a community-driven marketplace where AI developers, vibe coders, and power users can discover, share, and sell Agent Skills — the modular instruction packages that extend Claude and other AI platforms with specialized, repeatable workflows.

What is an Agent Skill?

A Skill is a folder containing a SKILL.md file with YAML frontmatter (name + description) and markdown instructions. Skills can include scripts, resources, and examples. They are packaged as ZIPs and loaded into AI systems to give them domain-specific expertise on demand.

Core Value Propositions

For Consumers

- Browse a curated library of skills instead of building from scratch
- Preview skill outputs with live examples before downloading
- Export guides for Claude.ai, Claude Code, and the Anthropic API
- Search by category, rating, use case, or AI compatibility

For Creators

- Publish free or paid skills to a global audience
- Showcase demo outputs to prove skill quality
- Earn revenue through one-time purchases or subscriptions
- Build reputation through ratings, downloads, and reviews

2. Core Feature Set

MVP Features (Phase 1)

Feature	Description	Priority
Skill Listings	Title, description, category, tags, author, version, compatibility	P0
Upload & ZIP Packaging	Validated SKILL.md upload with ZIP bundling	P0
Output Previews	Static screenshot or text block showing real skill output	P0
Search & Filters	Full-text search, category filter, free/paid toggle	P0
Download / Purchase	Free download or Stripe-gated paid skill	P0
User Profiles	Public profiles with published skills and stats	P1
Ratings & Reviews	Star ratings with written reviews	P1
Export Guides	Step-by-step instructions for Claude.ai, Claude Code, API	P1
Skill Validator	Auto-check SKILL.md frontmatter on upload	P1

Growth Features (Phase 2)

Feature	Description
Live Demo Sandbox	Run skill against Claude API in-browser to see live output
Skill Collections	Curated bundles (e.g. 'Dev Toolkit', 'Content Creator Pack')
Version Control	Skill versioning with changelog and diff viewer
Creator Dashboard	Analytics: downloads, revenue, reviews, impressions
Skill Forking	Fork and remix public skills with attribution
Compatibility Badges	Verified badges for Claude.ai / Claude Code / API
Newsletter / Digest	Weekly digest of top new and trending skills

3. Tech Stack Options

Three stack options are presented, ordered from fastest-to-ship to most scalable. Given your CS background, Option A is the recommended choice for speed without sacrificing control.

★ Option A — Recommended: T3 Stack (TypeScript Full-Stack)

Best for: fast shipping, type safety, strong community, easy Stripe + auth integration

Layer	Technology	Why
Framework	Next.js 14 (App Router)	File-based routing, SSR/SSG, API routes — one repo for everything
Language	TypeScript	Catch bugs at compile time, great DX with autocomplete
Styling	Tailwind CSS + shadcn/ui	Rapid UI with accessible pre-built components
Database	PostgreSQL (Supabase or Neon)	Relational, great for skills + users + reviews + transactions
ORM	Prisma	Type-safe DB queries, schema migrations, auto-generated types
Auth	Clerk or NextAuth.js	GitHub/Google OAuth, magic links — 30min setup
File Storage	Supabase Storage or AWS S3	ZIP uploads, README previews, screenshots
Payments	Stripe Connect	Marketplace payouts to creators, split fees, subscriptions
Search	Postgres Full-Text + Meilisearch	Start with PG FTS, add Meilisearch when needed
Deployment	Vercel	Zero-config Next.js deployment, edge functions
CI/CD	GitHub Actions	Auto-deploy on push, lint, test

Option B — Decoupled: React + FastAPI

Best for: if you plan to expose a public API from day one or want Python on the backend

Layer	Technology
Frontend	React + Vite + Tailwind CSS
Backend	FastAPI (Python)
Database	PostgreSQL + SQLAlchemy / Alembic
Auth	FastAPI-Users or Auth0

File Storage	AWS S3 or Cloudflare R2
Payments	Stripe Connect
Search	Elasticsearch or Meilisearch
Deployment	Frontend: Vercel / Netlify · Backend: Railway, Fly.io, or AWS ECS

Option C — Low-Config: Next.js + PlanetScale + Uploadthing

Best for: absolute quickest launch, smallest surface area, serverless-first

Layer	Technology
Framework	Next.js 14
Database	PlanetScale (MySQL, serverless)
ORM	Drizzle ORM
Auth	NextAuth.js
File Upload	Uploadthing (simplest DX for ZIPs + images)
Payments	Stripe
Deployment	Vercel

Shared Essentials Across All Options

- Skill validator — parse & lint SKILL.md frontmatter server-side on upload (js-yaml or PyYAML)
- ZIP handling — adm-zip (Node) or zipfile (Python) to inspect & serve skill packages
- Email — Resend or SendGrid for transactional emails (purchase receipts, new reviews)
- Analytics — PostHog (open-source, privacy-friendly) for usage tracking
- Rate limiting — Upstash Redis for API rate limiting

4. Core Data Model

Entity	Key Fields
User	id, email, username, bio, avatar, stripe_account_id, created_at
Skill	id, author_id, name, slug, description, category, tags[], price, download_count, zip_url, preview_images[], skill_md_content, ai_compatibility[], version, created_at

Review	id, skill_id, user_id, rating (1-5), body, created_at
Purchase	id, buyer_id, skill_id, stripe_payment_intent_id, amount, created_at
Collection	id, curator_id, title, description, skill_ids[]

5. Action Plan & Roadmap

Phase 0 — Foundation (Week 1–2)

- Set up repo, CI/CD pipeline, and project structure (Next.js monorepo)
- Configure Supabase (DB + Storage) and run initial Prisma migrations
- Integrate Clerk/NextAuth for GitHub + Google OAuth
- Deploy skeleton app to Vercel with staging environment
- Define full Prisma schema: User, Skill, Review, Purchase

Phase 1 — Core Marketplace (Week 3–6)

- Build skill upload flow: SKILL.md validation, ZIP bundling, S3/Supabase upload
- Build skill listing page with search, filters, pagination
- Build skill detail page: preview images, description, export guide, download button
- Implement free downloads (auth-gated, download count tracking)
- Public user profiles with published skills grid
- Add ratings + reviews system

Phase 2 — Monetization (Week 7–9)

- Integrate Stripe Connect for creator onboarding and marketplace payouts
- Paid skill purchase flow with Stripe Checkout + webhook handling
- Creator earnings dashboard (revenue, payouts, download stats)
- Promo codes and discount system

Phase 3 — Growth & Community (Week 10–14)

- Live demo sandbox: run skill against Claude API in-browser preview
- Skill Collections / Bundles curated by staff and top creators
- Skill forking with attribution chain
- Compatibility verification badges (Claude.ai / Claude Code / API)
- Email digest: weekly trending skills newsletter via Resend
- SEO optimization: skill pages as statically generated, sitemap, OG images

Phase 4 — Scale (Week 15+)

- Meiliseach integration for instant, typo-tolerant search at scale
- Skill versioning with changelog and diff view
- API for programmatic skill access (for power users / integrations)
- Partner program: highlight partner skills from companies like Notion, Figma
- Mobile-responsive PWA polish

Key Milestones

Milestone	Target	Success Metric
Private Beta	Week 6	10 skills uploaded, core flow working end-to-end
Public Launch	Week 9	50+ skills, Stripe payments live, social share
Community Growth	Week 14	200+ skills, 500+ registered users, first paid sales
Scale	Week 20	1,000+ skills, \$1k MRR from paid skills

6. Monetization Model

Model	How It Works	Revenue Split
Paid Skills (one-time)	Creator sets price, buyer pays via Stripe	Creator 80% / Platform 20%
Skill Subscriptions	Monthly access pass to creator's skill library	Creator 80% / Platform 20%
Featured Listings	Paid promotion to appear in 'Featured' section	Platform 100%
Pro Creator Plan	Unlimited uploads, analytics, priority support	Platform 100% (SaaS)
Enterprise Licensing	Team/org bulk skill access with SSO	Platform 100% (B2B SaaS)

7. Competitive Edge & Risks

Differentiators

- First dedicated marketplace for Agent Skills — no direct competitors today
- Live demo sandbox proves skill quality before purchase — highest trust signal

- Open Agent Skills standard (agentskills.io) means cross-platform potential beyond Claude
- Community forking & remixing drives organic content growth

Risks & Mitigations

Risk	Mitigation
Anthropic builds this themselves	Stay platform-agnostic (agentskills.io spec); expand to other AI platforms
Low-quality skill spam	Validator on upload + community flagging + manual curation for featured
Chicken-and-egg supply problem	Seed with 20–30 high-quality skills yourself on launch day
Piracy of paid skills	Watermarked ZIPs, DMCA process, license enforcement in ToS
API cost for demo sandbox	Rate-limit demos per user per day; add cost to Pro plan